

Machine Learning Syllabus

Module 1: Introduction to Machine Learning

- Applications of Machine Learning
- Difference between Traditional Coding and Machine Learning
- Supervised vs Unsupervised vs Reinforcement Learning
- Types of Machine Learning: Regression and Classification

Module 2: Simple Linear Regression

- Simple Linear Regression
- Machine Learning Workflow

Module 3: Python Basics

- Python
- NumPy, Pandas
- Train-Test Split
- Overfitting and Underfitting Basic Idea

Module 4: Model Training, Validation, and Tuning

- Training Machine Learning Model: Linear Regression
- Train-Test Split
- Hyperparameter Tuning
- Regression Metrics

Module 5: Data Preprocessing and EDA

- Task 1
- Data Preprocessing
- Data Visualization
- Exploratory Data Analysis

Module 6: Feature Engineering

- Feature Engineering
- Feature Scaling
- Transformers
- Outliers
- Missing Values Handling
- Encoding
- Duplicate Value Handling
- Binning

- Date/Time Feature Extraction
- ColumnTransformer
- Task 2

Module 7: Mathematics and Optimization

- Integration and Differentiation
- Loss Function
- Gradient Descent and its Types
- Learning Rate
- Task 3

Module 8: Advanced Regression Concepts

- Bias-Variance Trade-off
- Multiple Linear Regression
- Polynomial Regression
- Task 4

Module 9: Classification

- Logistic Regression
- Classification Metrics
- Imbalanced Dataset
- Softmax Regression
- Cross-Validation
- K-Fold Cross-Validation
- Project 1

Module 10: Linear Algebra and Statistics

- Linear Algebra
- Probability
- Statistics

Module 11: Linear Regression Assumptions

- Assumptions of Linear Regression

Module 12: Dimensionality Reduction

- Curse of Dimensionality
- PCA
- Feature Selection
- Task 5

Module 13: Classical Machine Learning Algorithms

- SVM
- KNN
- K-Means Clustering
- Clustering Evaluation
- Task 6

Module 14: Tree-Based and Ensemble Learning

- Decision Trees and Regression Trees
- Voting
- Bagging
- Random Forest
- Boosting
- XGBoost
- Stacking
- Task 7

Module 15: Regularization Techniques

- Regularization
- Ridge Regression
- Lasso Regression
- Elastic Net

Module 16: Introduction to Deep Learning

Module 17: Major Project and Evaluation for Internship